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Week 6, Git Hands On – 2

Objectives

- Explain branching and merging
- Explain about creating a branch request in GitLab
- Explain about creating a merge request in GitLab

In this hands-on lab, you will learn how to:

- Construct a branch, do some changes in the branch, and merge it with master (or trunk)

Prerequisites

The following are the pre-requisites to complete this hands-on lab:

- Setting up Git environment with P4Merge tool for Windows

Notes*:

Please follow the below steps for creating a free account in GitHub.

Do not use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **30 minutes.**

Please follow the instruction to complete the hands-on. Each instruction expects a command for the Git Bash.

Branching:

1. Create a new branch **"GitNewBranch"**.
2. List all the local and remote branches available in the current trunk. Observe the **"*"** mark which denote the current pointing branch.
3. Switch to the newly created branch. Add some files to it with some contents.
1. Commit the changes to the branch.
2. Check the status with **"git status"** command.

Merging:

1. Switch to the master
2. List out all the differences between trunk and branch. These provide the differences in command line interface.
3. List out all the visual differences between master and branch using **P4Merge tool**.
4. Merge the source branch to the trunk.
5. Observe the logging after merging using **"git log --oneline --graph --decorate"**
6. Delete the branch after merging with the trunk and observe the git status.

Solutions :

```
Windows PowerShell
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PS D:\Projects\Cognizant\git\gitDemo> git branch GitNewBranch
fatal: a branch named 'GitNewBranch' already exists
PS D:\Projects\Cognizant\git\gitDemo> git branch GitNewBranch1
PS D:\Projects\Cognizant\git\gitDemo> git branch -a
  GitNewBranch
  GitNewBranch1
* master
```

```
PS D:\Projects\Cognizant\git\gitDemo> git commit -m "Added sample file in GitNewBranch"
[GitNewBranch 889e356] Added sample file in GitNewBranch
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 sample1.txt
PS D:\Projects\Cognizant\git\gitDemo> git status
On branch GitNewBranch
nothing to commit, working tree clean
PS D:\Projects\Cognizant\git\gitDemo> git checkout master
Switched to branch 'master'
PS D:\Projects\Cognizant\git\gitDemo> git merge GitNewBranch
Updating c271fbd..889e356
Fast-forward
 .gitignore | Bin 0 -> 32 bytes
 sample.txt | Bin 0 -> 56 bytes
 sample1.txt | Bin 0 -> 56 bytes
3 files changed, 0 insertions(+), 0 deletions(-)
```

```
create mode 100644 .gitignore
create mode 100644 sample.txt
create mode 100644 sample1.txt
PS D:\Projects\Cognizant\git\gitDemo> git log --oneline --graph --decorate
* 889e356 (HEAD -> master, GitNewBranch) Added sample file in GitNewBranch
* 2121fb0 Added sample file in GitNewBranch
* c271fbd (GitNewBranch1) initial commit
PS D:\Projects\Cognizant\git\gitDemo> git branch -d GitNewBranch
Deleted branch GitNewBranch (was 889e356).
PS D:\Projects\Cognizant\git\gitDemo> git status
On branch master
nothing to commit, working tree clean
PS D:\Projects\Cognizant\git\gitDemo> |
```

```
PS D:\Projects\Cognizant\git\gitDemo> git checkout GitNewBranch
Switched to branch 'GitNewBranch'
PS D:\Projects\Cognizant\git\gitDemo> echo This is a sample file > sample1.txt
PS D:\Projects\Cognizant\git\gitDemo> git add .
```